

Principles of Economics

Class 10: Economic Fluctuations and Unemployment

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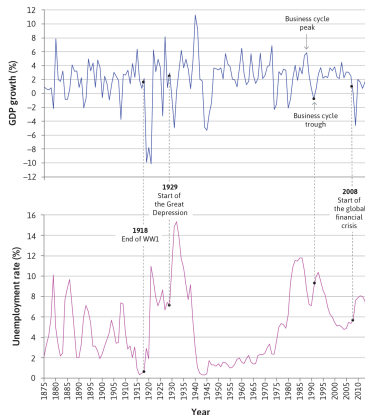


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- ▶ In the last class, we analysed how **people make decisions about saving and consumption** (Micro, Topic 9):
 - ▶ We focused on the role played by **consumer preferences** (impatience and consumption smoothing);
 - ▶ And the **interest rates**.
- ▶ In any case, these decisions also depend on current and expected **economic conditions**:
 - ▶ The **price level**, or **unemployment** affect consumer behaviour:
An increase in prices → anticipated consumption / An increase in unemployment → reduction in consumption.
 - ▶ Therefore, **consumers' decisions** affect **firms' decisions**:
If firms expect consumption to fall, they will normally cut their production and investment levels.
- ▶ Today we will focus on **three aspects**:
 - ▶ **Which indicators** are used to measure economic conditions?
 - ▶ How (and why) do **economic conditions** relate to **consumption and investment decisions**?
 - ▶ How does the business cycle relate to **inflation and unemployment**?

1. Introduction: The Business Cycle

- ▶ **Economic growth is not a stable process:**
- ▶ Positive and negative growth rates alternate;
- ▶ These fluctuations are called **business cycles**.
- ▶ The cycle has **two important periods**:
 - ▶ **Expansion**: output rises, or sits above potential.
 - ▶ **Recession phases**: output decreases or is below its potential. (The definition of “potential” will be seen later)



Introduction: The Business Cycle

Measuring the Aggregate Economy

Economic Fluctuations and Consumption

Economic fluctuations and investment

Economic fluctuations and...
...other components of GDP

Economic fluctuations and inflation

Economic fluctuations and unemployment

Summary

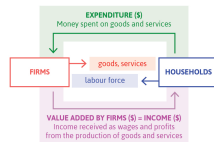
2. Measuring the Aggregate Economy

Measuring the aggregate economy

Definition of GDP

- ▶ First, how to measure economic activity:
 - ▶ **Gross Domestic Product (GDP)** is the most widely used measure
 - ▶ GDP comes from a **system of national accounts**:
 - ▶ Standards for measuring production and expenditure.
- ▶ There are **3 equivalent ways to measure GDP**:

Method	Definition	Focused on:
Expenditure	Total spending on domestic products	Final products sold in the economy;
Production	Total domestic output measured in aggregate terms	Increase in the value of products in production.
Income	Total domestic income	Income generated (wages, interest, profits, rents).

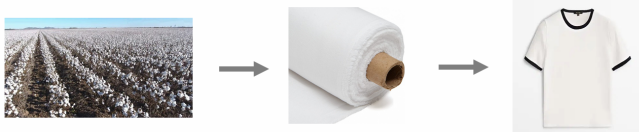


* The simplified circular flow model shows the equivalence: $Y = C + I$,
Y: output; C: consumption, I: investment

Measuring the aggregate economy

Definition of GDP

- ▶ A small economy that produces only t-shirts:



Method	Cotton plantation	Cotton fabric production	Cotton t-shirt production	GDP
Sales	50€	80€	100€	-
Value added	50€	$30€ = 80€ - 50€$	$20€ = 100€ - 80€$	100€
Expenditure (final)				100€
Income	Wages: 25€ Profits: 25€	Wages: 15€ Profits: 15€	Wages: 10€ Profits: 10€	100€

- ▶ Same GDP with any of the three methods.

- ▶ Aggregate demand is GDP seen from the expenditure side:

$$Y = \text{Consumption} + \text{Investment}$$

- ▶ And **government spending and trade?**

- ▶ **International trade:**

- ▶ **Imports:** consumed here, produced abroad
- ▶ **Exports:** produced here, consumed abroad

GDP measures **domestic production**: *imports out, exports in*.

- ▶ **Government activity:**

Taxes finance two very different things:

- ▶ **Government spending:** measured at its cost of production, taken as its value added.
- ▶ **Transfers:** they move value, they do not create it, so they do not count

Measuring the aggregate economy

Components of GDP

- By the expenditure method:

$$GDP = Y = C + I + G + X - M$$

- **Output/Aggregate Demand (Y):** Domestic production
 - **Consumption (C):** spending on goods and services
 - **Investment (I):** new capital goods, plus the change in inventories
 - **Government spending (G):** goods and services for public provision
 - **Net exports (X-M):** Exports minus imports
- Which component weighs most?

GDP Components (in %, in 2024)							
Country	C	G	I	Inventories	X-M	X	M
US	67,9	17,2	18	0,2	-3,1	11	14,1
EU19	52,1	21,5	21,2	0	+4,3	50,4	46,1
China	40	17,2	41	0	+2,8	20	17,2

- It depends on the economic structure.

Measuring the aggregate economy

Measurement problems

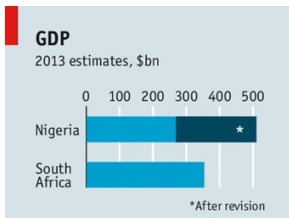
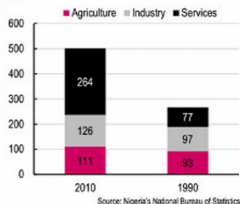


Figure 1: Nigeria – 2010 series vs 1990 GDP series, \$bn



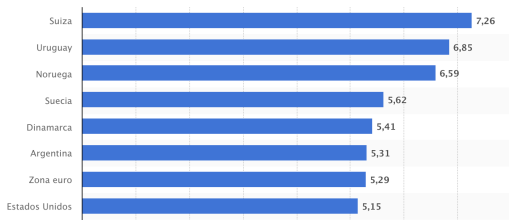
- ▶ The measurement of GDP is heavily influenced by certain assumptions:
- ▶ “Nigeria overtook South Africa on Sunday, becoming the **largest economy in Africa** and the 26th in the world.”
- ▶ “Nigeria’s GDP in 2013 was 509 billion US dollars, **89% higher than previously established**”
- ▶ “The change lies in the **updating of the base year for its calculation from 1990 to 2010, when the structure of the economy was still very different** and services such as banking and telecommunications were barely taken into account.”
- ▶ Read the [full article](#) in the Financial Times.

Measuring the aggregate economy

Measurement problems

- ▶ **The measurement of GDP is heavily influenced by certain assumptions:**
- ▶ Different countries with different currencies...
...we could use the exchange rate but...
- ▶ Even countries with the same currency (euro area)...
...have different prices

Big Mac price (The Economist).



Purchasing Power Parity (PPP) tries to correct these differences.

Who contributes most to global growth?

The United States does, when market exchange rates are used ...
(percent of global GDP growth, 2006)



but China does, when PPP rates are used.
(percent of global GDP growth, 2006)



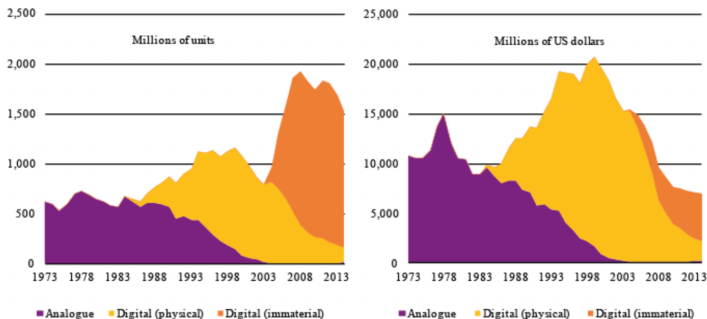
Source: IMF, World Economic Outlook database (September 2006).

Measurement problems

Other aspects:

- ▶ How to measure **digital products delivered** at zero price?
 - ▶ They are valued by consumers but excluded from GDP. For example, Wikipedia.
 - ▶ Read the [VoxEU article](#).

Figure 1. Comparison of units and value of US music sales



Other aspects:

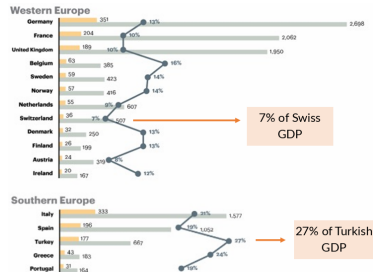
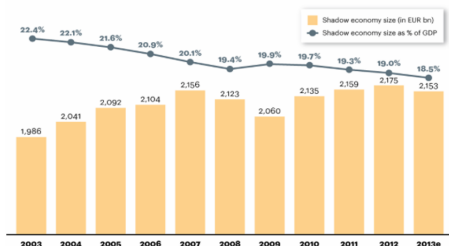
- ▶ How to measure **digital products delivered** at zero price?
 - ▶ They are valued by consumers but excluded from GDP. For example, Wikipedia.
 - ▶ Read the [VoxEU article](#).
- ▶ GDP does not account for **non-market activities**:
 - ▶ For example, environmental costs
 - ▶ The difficulty of their measurement has already been discussed, considering externalities. More on core-econ.
- ▶ GDP also does not include **unpaid household work (excluded from the market)**:
 - ▶ “If the value of non-market household work is incorporated, output raises nominal GDP by 39% in 1965 and by 26% in 2010.”
 - ▶ More in [Bridgman, Dugan, Lal, Osborne and Villones \(2012\)](#)

Measuring the aggregate economy

Measurement problems

- ▶ GDP also does not reflect the **shadow economy**:
 - ▶ The shadow economy covers all **activities that are not formally declared** (neither taxes nor regulation);
- ▶ How relevant is it in **Europe**?

Figure 2
The development of the shadow economy in Europe



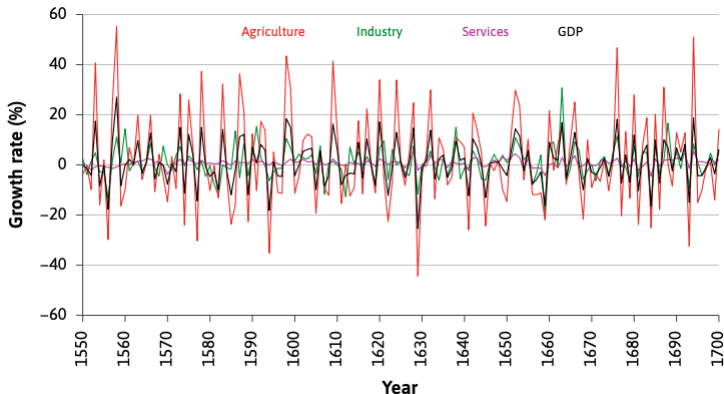
For more details, read the article by [Schneider \(2013\)](#) or by [Medina and Schneider \(2018\)](#).

3. Economic Fluctuations and Consumption

Shocks to economic activity

- ▶ Economic fluctuations between bad and good times in industrial societies and also in **agrarian** ones:

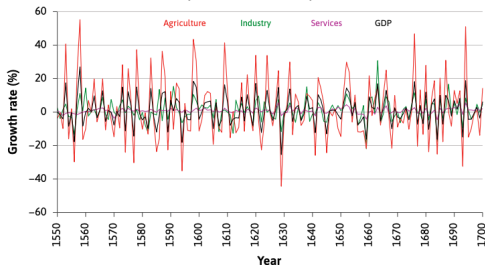
GDP fluctuations, total and by sector, in the British agrarian economy (1550 - 1700)



Shocks to economic activity

- ▶ Economic fluctuations between bad and good times in industrial societies and also in **agrarian** ones:
- ▶ At that time:
 - ▶ The agricultural sector was the most **volatile**;
 - ▶ **Climate variations** were key.
- ▶ **Shocks** are unexpected events that cause fluctuations in economic activity.

GDP fluctuations, total and by sector, in the British agrarian economy (1550 - 1700)



- ▶ Two **types of shock**:
 - ▶ **Positive**: technology, new resources (oil, lithium for batteries).
 - ▶ **Negative**: Covid-19, wars, debt crises, Brexit.
- ▶ Each component of GDP amplifies or dampens the shock. Starting with consumption.

- ▶ One **feature of households** explains everything that follows:
 - ▶ Households prefer a **stable level of consumption**.
- ▶ How do they absorb a **sudden cut in income**?
 - ▶ **Self-insurance**: their own saving and borrowing;
 - ▶ **Co-insurance**: family, networks and the government.
- ▶ **Co-insurance** is less effective when the negative shock affects everyone **at the same time**:
 - ▶ Which is exactly when it matters most.

- ▶ Households plan consumption on expectations about the future:
 - ▶ **They react to shocks and adapt** their plans;
 - ▶ And the reaction depends on how persistent the shock is:
 - ▶ **Temporary shocks**
 - should not alter consumption.
 - ▶ **Permanent shocks**
 - force an adjustment.

- ▶ Consumption smoothing **stabilises the economy**:
 - ▶ It keeps shocks from being **too sharp**;
 - ▶ Without it, **both negative and positive shocks would be larger**
 - ▶ And fluctuations are costly: unemployment, inflation.
- ▶ But **smoothing has limits**:
 - ▶ **Credit constraints, lack of willpower, or limited co-insurance**;
 - ▶ So it is **often not enough to stabilise the economy**.
- ▶ Those limits are what the rest of the class is about.

Limits to consumption: credit constraints

- ▶ **Credit constraints bite:** consumption stays low after a banking crisis.
- ▶ The fall after 2008 is telling.
- ▶ Read more in the [VoxEU article](#).

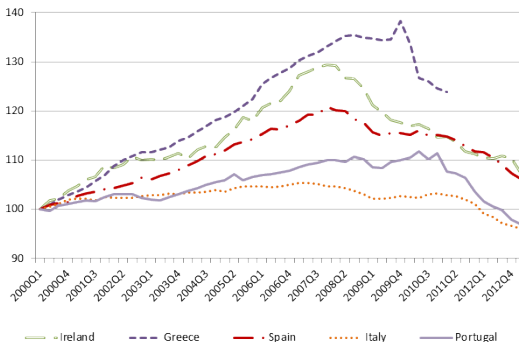


Figure 1. Real per-capita consumption spending (2000 first quarter = 100) Source: Petra Gerlach-Kristen, Rossana Merola, Conor O'Toole, VoxEU, 01 December 2013

4. Economic fluctuations and investment

The volatility of investment

- ▶ **Firms do not want stability the way households do:**
 - ▶ They adjust investment to **every shock**, temporary or permanent.
- ▶ In general:
 - ▶ High demand → lack of capacity → high investment → higher demand.
- ▶ Expectations are decisive:
 - ▶ You add capacity if you expect demand to grow.
 - ▶ Otherwise, you will not invest.



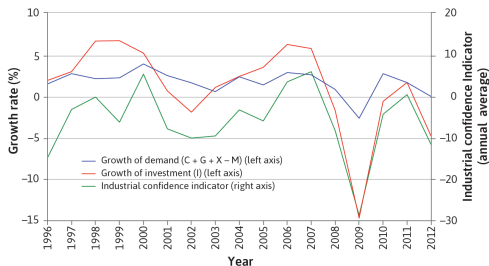
Investment coordination

- ▶ Virtuous and vicious investment cycles are a **coordination problem**.
- ▶ **Investment game:**
 - ▶ Two players: Firms A and B
 - ▶ Actions: Invest or not invest;
 - ▶ Structure: They decide simultaneously;
 - ▶ Payoffs: profits from investment.
- ▶ **Two Nash equilibria:**
 - ▶ (100,100) and both invest;
 - ▶ (10,10) and neither invests;
 - ▶ Coordinating would reach the best outcome, social and individual.
- ▶ **Investing pays off** when the others invest too.

		B's profit	
		B invests	B does not invest
A's profit	A invests	100	80
	A does not invest	-40	10

Investment coordination

- ▶ Firms do not coordinate **one to one**:
 - ▶ Statistical institutes publish **industrial confidence indicators** from surveys;
 - ▶ And **firms read them** to guess what the others will do;
- ▶ There is a **strong correlation between investment and industrial confidence indicators**.



UK small business confidence falls to lowest level since pandemic

Outlook darkens as companies are squeezed by rising costs and falling revenues



Nine in ten small businesses complain of the increasing burden of energy and wage bills © Matthew Ashmore/Alamy

Daniel Thomas and Peter Foster OCTOBER 19 2022



Confidence among small British business owners has fallen to its lowest level since the depths of the coronavirus pandemic as they are squeezed by rising costs and falling revenues, according to the Federation of Small Businesses.

Read the [article from the FT](#)

German recession looms as business confidence slumps to two-year low

Economists warn of downturn as high prices and threat of gas shortage weigh on economy



Berlin last month triggered the second stage of its national gas emergency plan, a move that brought the German government a step closer to rationing supplies © Clemens Bilan/EPA-EFE/Shutterstock

Martin Arnold in Frankfurt JULY 25 2022

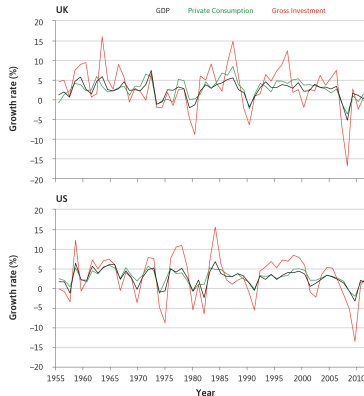


German business confidence has fallen to its lowest level for more than two years in the latest sign that Europe's largest economy is teetering on the brink of recession.

Read the [article from the FT](#)

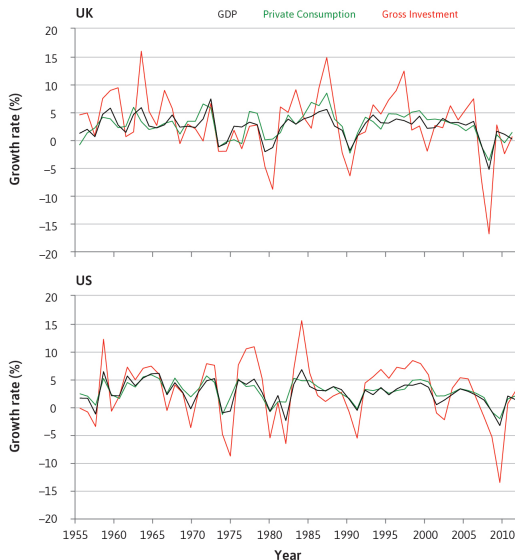
Investment and the aggregate economy

- ▶ So **investment tracks the business cycle**.
- ▶ Moreover, it **amplifies the cycle**:
 - ▶ If investment increases, you also invest more:
→ **the expansion is amplified**;
 - ▶ If investment decreases, you also invest less:
→ **the recession is amplified**.
 - ▶ **Coordination self-reinforces the cycles**.
- ▶ **Investment is normally more volatile than GDP**;
- ▶ **Smoothing keeps private consumption steadier**.



Economic fluctuations and investment

Investment and the aggregate economy



5. Economic fluctuations and... ...other components of GDP

- ▶ The behaviour of **NET EXPORTS** also affects GDP fluctuations:
 - ▶ **Exports** follow foreign demand, that is, the cycle of the markets you sell to:
 - ▶ Hence the synchronised cycles of integrated countries, like the EU;
 - ▶ **Imports** depend on the domestic economy.
- ▶ **GOVERNMENT SPENDING** is typically less volatile than investment:
 - ▶ **Countercyclical policy**
 - ▶ **Expansions:** *Decrease* spending to reduce the expansion
 - ▶ **Recessions:** *Increase* spending to reduce the recession.
 - ▶ **Procyclical policy**
 - ▶ **Expansions:** *Increase* spending to increase the expansion.
 - ▶ **Recessions:** *Decrease* spending to increase the recession.
- ▶ And it can turn **contractionary for budgetary reasons**, even in a recession [austerity, 2010].

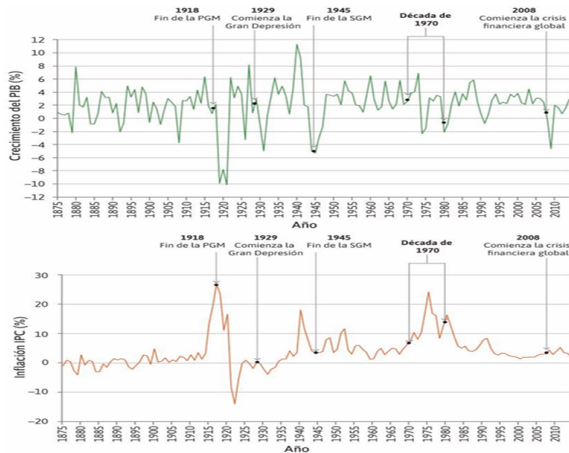
6. Economic fluctuations and inflation

- ▶ **Inflation**: a rise in the general price level.
- ▶ Inflation **rises with expansions** (higher aggregate demand)...
...And **slows down in recessionary periods** (lower aggregate demand)
- ▶ Inflation shows a relationship:
 - ▶ **positive with GDP growth;**
 - ▶ **negative with unemployment.**

Economic fluctuations and inflation

Introduction

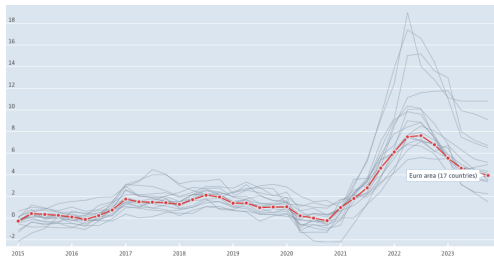
- ▶ Inflation shows a relationship:
- ▶ **positive with GDP growth;**
- ▶ **negative with unemployment.**



UK data.

The euro area

- ▶ Inflation rose **after the pandemic**, in the euro area:
 - ▶ The pandemic broke **supply chains and shifted consumption**;
 - ▶ When restrictions eased, **consumption rebounded and production could not follow**;
 - ▶ **The war in Ukraine made it worse**: energy costs, disrupted activity.



IMF report [World Economic Outlook 2022](#).

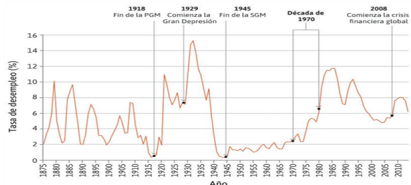
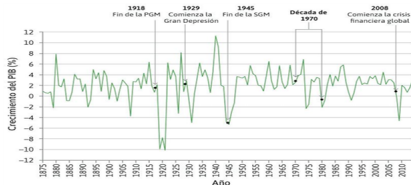
7. Economic fluctuations and unemployment

- **Okun's Law:** a strong, stable **negative relationship between unemployment and GDP growth.**



Art Okun (1928-1980)

- **Main idea:**
 - Output falls
 - Firms need fewer workers
 - Unemployment rises



- ▶ **Okun's Law:** a strong, stable **negative relationship between unemployment and GDP growth.**

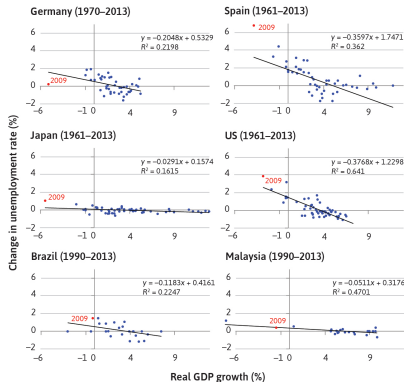


Art Okun (1928-1980)

- ▶ Okun investigated the **relationship between GDP growth and unemployment:**

- ▶ A fall in the GDP growth rate of 1% is associated with an increase in unemployment of 0.5%.

- ▶ This negative relationship has been demonstrated empirically.
Okun coefficient = Degree of correlation



8. Summary

Section by section (Macro, Topic 3)

1. **The business cycle** → growth is not a steady process: expansions and recessions around potential output.
2. **Measuring the aggregate economy** → GDP by expenditure, production or income gives the same figure; $Y = C + I + G + X - M$.
3. **Measurement problems** → prices, PPP, free digital goods and the shadow economy: what GDP leaves out.
4. **Fluctuations and consumption** → households smooth, and that stabilises; credit constraints are the limit.
5. **Fluctuations and investment** → firms do not smooth: a coordination game that amplifies the cycle.
6. **Other components of GDP** → exports synchronise cycles; public spending can be counter- or pro-cyclical.
7. **Inflation and unemployment** → inflation goes with the cycle, unemployment against it: Okun's Law.

Key concepts: business cycle, expansion and recession, potential output, GDP by expenditure, production and income, value added, PPP, shadow economy, shock, consumption smoothing, self- and co-insurance, credit constraints, coordination game, business confidence, counter- and pro-cyclical policy, inflation, Okun's Law.